

Mahmoud Alyosify

Specialised Data Scientist | Predictive Modelling, Experimentation & Decision Analytics | MSc Artificial Intelligence, Queen's University (Canada)

✉ mahmoud.alyosify@gmail.com | 📞 +20 114 555 7959 | 📍 New Cairo, Egypt · Open to Relocation
🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

Data Scientist who builds models around the **decision they have to support**, not around the leaderboard. Built a satellite collision-risk model deliberately restricted to data available **48+ hours** before closest approach — matching the real operational window — and a reproducible **repository-mining study** answering a measurable question about engineering effort. Strong statistical foundation: five years teaching **probability, statistics, discrete mathematics and graph data mining** to **10,000+ learners**, and research-grade experimentation including **paired non-inferiority testing** and controlled ablation sweeps. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**. **Microsoft Certified: Data Analyst Associate**.

EDUCATION

- MSc in Artificial Intelligence**
Queen's University, Kingston, Ontario, Canada

Sep 2025 – Nov 2026
Expected Nov 2026

 - **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
 - Coursework: **Deep Learning, Large Language Models, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines**.
- B.Sc. Computer & Information Science (Bioinformatics)**
Assiut University, Egypt

Sep 2019 – Jul 2023
GPA: 3.53 / 4.00

 - Quantitative foundation in statistics, probability, discrete mathematics, data analysis and bioinformatics.
 - Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Statistics & Experimentation	Probability & Statistical Inference, Hypothesis Testing, Paired Non-Inferiority Testing, Controlled Experiments & Ablation Design, Class-Imbalance Handling, Cross-Validation, Error Analysis, Reproducible Analysis
Modelling	Scikit-learn, XGBoost, LightGBM, Logistic & Linear Models, Feature Engineering, Hyperparameter Optimisation (Optuna), Model Explainability (SHAP), Deep Learning (PyTorch, TensorFlow)
Data Engineering	SQL, Pandas, NumPy, PySpark, ETL Pipelines, Data Cleaning & Validation, Repository & API Data Mining, MySQL, MongoDB, SQL Server
Analytics & Reporting	Power BI (<i>Microsoft Certified: Data Analyst Associate</i>), Matplotlib, Streamlit Dashboards, Crystal Reports, Executive-Ready Insight Communication
Deployment	AWS, Docker, FastAPI, ONNX, Git/GitHub, Linux, CI/CD
Programming	Python, SQL, R, C++, C#, MATLAB, Bash

SELECTED DATA SCIENCE PROJECTS

- SCRAP — Satellite Collision Risk Assessment & Prediction**
Python · Scikit-learn · Feature Engineering · Imbalanced Classification

2026

 - Built a supervised risk model constrained to conjunction data available **48+ hours** before closest approach — deliberately trading offline accuracy for a prediction that arrives in time to act on.
 - Engineered features from raw conjunction records and handled a severely imbalanced target, selecting evaluation metrics appropriate to rare, high-consequence events rather than headline accuracy.
- Analyzing Review Effort: Human vs. Agentic Pull Requests**
Data Mining · Statistical Analysis · Reproducible Pipelines

2026

 - Built an end-to-end **GitHub-mining pipeline** — collection, cleaning, cohort definition and analysis — to quantify whether AI-generated pull requests shift reviewer effort compared with human-authored ones.
 - Defined comparable cohorts and explicit inclusion criteria so the comparison measured the intended effect rather than sampling artefacts.
- SimCLR-Vision-SSL — Controlled 42-Experiment Study**
PyTorch · Experimental Design · Ablation Analysis

2026

 - Designed and executed a **42-experiment sweep** isolating which augmentations drive representation quality, reaching **84.30% top-1 accuracy** — a controlled study rather than a single tuned run.
- Verbosity-Aware Decoding — Research-Grade Evaluation Design**
MSc Research Project (CISC 898), Queen's University · Statistical Testing

2026 – Present

 - Built a **pre-registered evaluation protocol** using **paired non-inferiority testing** to prove a cost reduction did not degrade quality — the statistically correct framing for "no worse than", which a standard significance test cannot answer.

Additional Projects

- **ATM Transaction Analysis** — exploratory data analysis and statistical reporting on transaction data in **R**.
- **Diabetes Onset Prediction** — supervised classification pipeline on clinical risk-factor data.
- **Horus-OSINT** — curated **159,826** GTD/GDELT records into a training corpus and deployed a fine-tuned model on AWS for natural-language querying of global event data.
- **Also:** RL-Cloud-Autoscaler (PPO/DQN cost optimisation) · AudioShield · HORUS Sentinel · Vitalism Solution.

PROFESSIONAL EXPERIENCE

Machine Learning Instructor
National Telecommunication Institute (NTI)

Jul 2025 – Sep 2025
Cairo, Egypt

- Delivered a **90-hour** hands-on program covering supervised learning, **PCA** and neural networks, guiding students through complete analyses on real datasets.

Systems Engineer (Military Service)
National Company for Roads Building & Development

Feb 2024 – Mar 2025
Cairo, Egypt

- Supported backend data systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**.
- Co-developed an internal **data-management and reporting** application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**, delivering operational reporting across multiple departments.

Online Instructor — Statistics, Probability & Data Science
Udemy & YouTube

2020 – Present
Remote

- Authored and taught Arabic-language courses in **probability, statistics, discrete mathematics, algorithms and graph data mining** for **10,000+ learners (700K+ views)**.

AI in Cybersecurity Intern
e& Egypt — via NTI (On-the-Job Training)

Jun 2025 – Jul 2025
Cairo, Egypt

- Applied AI-driven analytical tooling across threat intelligence, digital forensics and enterprise security analysis.

CERTIFICATIONS & AWARDS

Microsoft Certified: Data Analyst Associate (Power BI)	AWS Academy — Data Engineering, ML Foundations, ML for NLP
Machine Learning Specialization — DeepLearning.AI	Data Analysis in R — 57357 Children’s Cancer Hospital
HCIP – AI (Huawei)	ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)
1st Place — Smart Cities Hackathon 2022	Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.